Group Members: 1. Jeremy Zhao 2. Vincent Chen 3. Armando Martinez

Problem 1

Solution

$$A \times B = \{(\text{Alex}, \text{Aanya}), (\text{Alex}, \text{Encong}), (\text{Aanya}, \text{Aanya}), (\text{Aanya}, \text{Encong})\}$$

$$\begin{aligned} P(A \times B) = \{\, &\emptyset, \\ &\{(\text{Alex}, \text{Aanya})\}, \{(\text{Alex}, \text{Encong})\}, \{(\text{Aanya}, \text{Aanya})\}, \{(\text{Aanya}, \text{Encong})\}, \\ &\{(\text{Alex}, \text{Aanya}), (\text{Alex}, \text{Encong})\}, \\ &\{(\text{Alex}, \text{Aanya}), (\text{Aanya}, \text{Aanya})\}, \\ &\{(\text{Alex}, \text{Aanya}), (\text{Aanya}, \text{Encong})\}, \\ &\{(\text{Alex}, \text{Encong}), (\text{Aanya}, \text{Aanya})\}, \\ &\{(\text{Alex}, \text{Encong}), (\text{Aanya}, \text{Encong})\}, \\ &\{(\text{Aanya}, \text{Aanya}), (\text{Aanya}, \text{Encong})\}, \\ &\{(\text{Alex}, \text{Aanya}), (\text{Alex}, \text{Encong}), (\text{Aanya}, \text{Aanya})\}, \\ &\{(\text{Alex}, \text{Aanya}), (\text{Alex}, \text{Encong}), (\text{Aanya}, \text{Encong})\}, \\ &\{(\text{Alex}, \text{Aanya}), (\text{Aanya}, \text{Aanya}), (\text{Aanya}, \text{Encong})\}, \\ &\{(\text{Alex}, \text{Encong}), (\text{Aanya}, \text{Aanya}), (\text{Aanya}, \text{Encong})\}, \\ &\{(\text{Alex}, \text{Aanya}), (\text{Alex}, \text{Encong}), (\text{Aanya}, \text{Aanya}), (\text{Aanya}, \text{Encong})\}\, \} \end{aligned}$$

If auditing Nickolas, let $A' = \{\text{Alex}, \text{Aanya}, \text{Nickolas}\}$ and $B' = \{\text{Aanya}, \text{Encong}, \text{Nickolas}\}$.

Thus

$$|P(A' \times B')| = 2^{|A' \times B'|} = 2^{|A'||B'|} = 2^{3 \cdot 3} = 2^9 = 512$$

Problem 2

Solution

An ordered pair belongs to both $A \times B$ and $B \times A$ exactly when both of its entries lie in $A \cap B$, so

$$(A \times B) \cap (B \times A) = (A \cap B) \times (A \cap B) \quad \Longrightarrow \quad |(A \times B) \cap (B \times A)| = |A \cap B|^2$$

By inclusion–exclusion, and since $|A \times B| = |A||B| = |B \times A|$,

$$|(A \times B) \cup (B \times A)| = |A||B| + |B||A| - |A \cap B|^2 = 2|A||B| - |A \cap B|^2$$

So the union has the same size as $A \times B$ (which has size $|A||B|$) exactly when

$$2|A||B| - |A \cap B|^2 = |A||B| \quad \Longleftrightarrow \quad |A \cap B|^2 = |A||B|$$

Situation 1: $A = B$

If $A = B$, then $A \cap B = A$, so $|A \cap B|^2 = |A|^2$ and $|A||B| = |A|^2$. The two sides are equal, so the condition holds. We can also see it directly: $B \times A = A \times B$, so

$$(A \times B) \cup (B \times A) = (A \times B) \cup (A \times B) = A \times B$$

Example: $A = B = \{\text{Alex}, \text{Aanya}\}$. Then $A \times B = B \times A$ has $2 \cdot 2 = 4$ elements, and the union also has 4 elements.

Situation 2: $A = \emptyset$ **(same argument for** $B = \emptyset$**)**

If $A = \emptyset$, then every ordered pair in $A \times B$ would need a first entry from A, and there is none, so $A \times B = \emptyset$. Likewise $B \times A = \emptyset$. Hence

$$(A \times B) \cup (B \times A) = \emptyset \cup \emptyset = \emptyset = A \times B$$

so both sides have size 0. In the formula this is the case $|A| = 0$, where $|A \cap B|^2 = 0 = |A||B|$ as well.

Example: $A = \emptyset$ and $B = \{\text{Alex}, \text{Encong}\}$. Then $|A \times B| = 0 \cdot 2 = 0$ and the union has size 0.

Problem 3

Solution

$$(A \oplus B) \setminus C$$

Problem 4

Solution

Statement 1:

False.

Both blocks were defined inside $\mathbb{Z} \times \mathbb{Z}$, so they contain elements that are not in S.

- $(0, 0)$ satisfies $a = -b$, but $0 \notin \mathbb{Z}^+$ and $0 \notin \mathbb{Z}^-$, so $(0, 0) \notin S$.
- $(2, 1)$ satisfies $|2| \neq |1|$, but both entries are positive, so $(2, 1) \notin S$.

Statement 2

True.

Contained in S: if $(a, b) \in X_1$ then $a \neq 0$ and $b = -a$, so the entries have opposite signs and $(a, b) \in S$. $X_2 \subseteq \mathbb{Z}^+ \times \mathbb{Z}^-$ and $X_3 \subseteq \mathbb{Z}^- \times \mathbb{Z}^+$, both by definition inside S.

Nonempty: $(1, -1) \in X_1$, $(1, -2) \in X_2$, $(-1, 2) \in X_3$.

Disjoint: X_2 and X_3 have opposite sign patterns, so they cannot share a pair. For $X_1 \cap X_2$: a pair in X_1 has $a + b = 0$, while X_2 requires $a + b \neq 0$. The same argument gives $X_1 \cap X_3 = \emptyset$.

Union = S: take any $(a, b) \in S$, so a and b are nonzero with opposite signs.

- If $a + b = 0$, then $a = -b$ and $(a, b) \neq (0, 0)$, so $(a, b) \in X_1$.
- If $a + b \neq 0$ and $a \in \mathbb{Z}^+$, then $b \in \mathbb{Z}^-$ and $(a, b) \in X_2$.
- If $a + b \neq 0$ and $a \in \mathbb{Z}^-$, then $b \in \mathbb{Z}^+$ and $(a, b) \in X_3$.

Every element of S lies in exactly one block, so X_1, X_2, X_3 form a partition of S.

Statement 3: $\mathbb{Z}^+ \times \mathbb{Z}^-$ and $\mathbb{Z}^- \times \mathbb{Z}^+$

True.

The blocks are nonempty ($(1, -1)$ and $(-1, 1)$) and disjoint since a first entry cannot be both positive and negative and thus a pair cannot lie in both products.

Statement 4: $\mathbb{Z}^+ \times \mathbb{Z}^+$ and $\mathbb{Z}^- \times \mathbb{Z}^-$

False.

For instance $(1, 1)$ is in the first block but not in S, and every element of S